

# LUNAR AUTONOMOUS LANDING ZONE ASSESSMENT

Professional User Guide | AI-powered survey, hazard detection, rover route planning, and habitat site analysis

### LUNAR AUTONOMOUS LANDING ZONE ASSESSMENT

AI-POWERED SURVEY | HAZARD DETECTION | HABITAT PLANNING

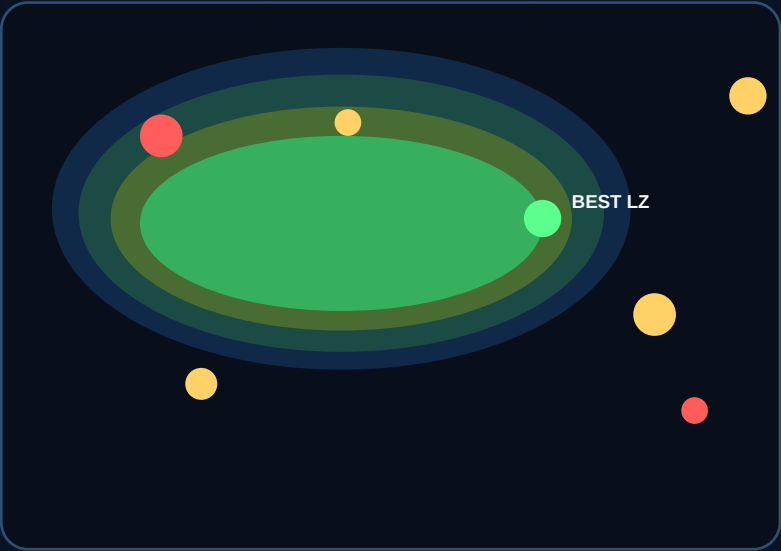
MISSION CONTROLS

Mission ID

Terrain Model

Scoring Weights

Rover Planning



MISSION OPS

Target Body: Moon

Profile: LZ + Habitat

System: **NOMINAL**

Risk: **LOW**

Craters: 12

Boulders: 28

PRIMARY FUNCTION

Assess terrain safety

Find candidate landing zones

Plan surface survey routes

CORE OUTPUTS

LZ score and risk band

Slope, solar, comm indices

JSON and CSV mission exports

MISSION CONTEXT

Lunar operations focus

Adaptable target body menu

Habitat preparation workflow

Autonomy | Intelligence | Exploration | Systems Engineering

LUNAR AUTONOMOUS LANDING ZONE ASSESSMENT | USER GUIDE

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# 1. PURPOSE AND OPERATING CONCEPT

This Streamlit application is a rapid mission-analysis tool for autonomous landing zone assessment. It creates a synthetic terrain field, evaluates local hazards, scores candidate touchdown zones, and generates a surface survey plan that supports habitat placement, power siting, and communication relay planning.

The app is framed around lunar operations, but it includes target body selections for Mars, Phobos, Europa, and custom airless bodies. The current terrain model is procedural and simulation-based, so it is best used for concept development, classroom demonstration, interface prototyping, and early systems engineering trade studies.

|                  |                                                                                                                                                        |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operational Goal | Help mission planners identify safer landing and habitat-preparation zones from terrain, slope, hazard, solar, and communications indicators.          |
| Primary User     | Student engineer, mission analyst, autonomy developer, or systems engineering team conducting preliminary LZ evaluation.                               |
| Decision Output  | Recommended landing zone coordinates, LZ score, risk band, local hazard count, rover survey path, habitat offset plan, and exportable mission summary. |

## Recommended workflow

- Set the mission ID, target body, mission profile, and terrain parameters in the sidebar.
- Tune hazard, solar, and communication scoring weights to match the mission priority.
- Run the autonomous assessment and review the LZ score, slope, local hazards, solar index, and communication index.
- Use the 3D terrain map and secondary analysis panels to validate the recommended site.
- Export JSON or CSV for reporting, traceability, and follow-on analysis.

## 2. SYSTEM ARCHITECTURE AT A GLANCE

The app is organized as a compact mission-analysis pipeline. Inputs flow from the sidebar into the procedural terrain model. The terrain model produces elevation, crater, and boulder fields. The scoring engine then converts those features into slope, hazard, solar, communication, and aggregate suitability layers.

| Layer              | Purpose                                          | Key Code Elements                                                                                                                                                    |
|--------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User Interface     | Collect mission and model parameters.            | Streamlit sidebar, metric cards, Plotly charts, download buttons                                                                                                     |
| Terrain Generator  | Create a repeatable synthetic lunar-like map.    | <code>generate_lunar_terrain()</code> , <code>seed</code> , <code>grid_size</code> , <code>roughness</code> , <code>crater_count</code> , <code>boulder_count</code> |
| Hazard Model       | Estimate crater and boulder risk.                | <code>radial_penalty()</code> , <code>hazard_score</code> , local hazard count                                                                                       |
| Suitability Engine | Calculate the best LZ based on weighted scoring. | <code>landing_suitability()</code> , <code>slope_score</code> , <code>solar_score</code> , <code>comm_score</code>                                                   |
| Planning Layer     | Create habitat offsets and rover survey paths.   | <code>rover_path()</code> , <code>path_distance()</code> , <code>plan_df</code>                                                                                      |
| Export Layer       | Package mission summary for reporting.           | JSON summary, CSV summary, <code>timestamp_utc</code>                                                                                                                |

### Architecture interpretation

This is a decision-support architecture, not a flight-certified autonomous landing system. The app demonstrates how a landing-zone assessment stack can combine perception-inspired hazard fields, mission-weighted scoring, and surface operations planning into one integrated dashboard.

### 3. MISSION CONTROLS

The sidebar controls define the scenario and determine how the simulation weighs competing mission needs. Use a fixed seed when you need repeatable results for reports, demonstrations, or validation screenshots.

| Control            | What It Does                                     | Practical Guidance                                                                    |
|--------------------|--------------------------------------------------|---------------------------------------------------------------------------------------|
| Mission ID         | Labels the scenario and file exports.            | Use a unique ID such as LZAP-2026-001 for traceability.                               |
| Target Body        | Sets the mission context shown in the dashboard. | Moon is the primary design case; other targets are selectable for concept comparison. |
| Mission Profile    | Defines the planning emphasis.                   | Use Landing Zone + Habitat Preparation for the full workflow.                         |
| Simulation Seed    | Controls repeatability.                          | Keep the same seed when comparing scoring-weight changes.                             |
| Map Resolution     | Controls grid detail.                            | Higher resolution looks smoother but can slow rendering.                              |
| Regolith Roughness | Adds terrain variability.                        | Higher values increase slope variation and reduce smooth LZ candidates.               |
| Crater Density     | Adds crater hazards.                             | Higher density increases exclusion pressure around candidate sites.                   |
| Boulder Density    | Adds point obstacles.                            | Higher density makes rover planning and touchdown risk more restrictive.              |

#### Weight controls

Hazard avoidance carries the highest mission relevance because touchdown safety depends on clear terrain, low slope, and adequate obstacle clearance. Solar and communication weights should be adjusted only after confirming that the landing site remains physically safe.

## 4. TERRAIN AND HAZARD MODEL

The terrain generator produces a synthetic elevation field using sinusoidal terrain features, random roughness, crater bowls with raised rims, and boulder-height perturbations. This creates a visually useful analog for lunar-like terrain, without claiming fidelity to a specific landing site.

| Feature        | Model Behavior                              | Operational Meaning                                              |
|----------------|---------------------------------------------|------------------------------------------------------------------|
| Base Elevation | Combines broad waves and noise.             | Represents rolling terrain and local regolith irregularity.      |
| Craters        | Adds depressed bowls and raised rims.       | Crater interiors, rims, and nearby gradients reduce suitability. |
| Boulders       | Adds local raised obstacles.                | Obstacle fields penalize touchdown and rover traverse safety.    |
| Slope          | Calculated from elevation gradients.        | Lower slope improves touchdown stability and site usability.     |
| Hazard Penalty | Radial penalty around craters and boulders. | Closer and larger hazards have stronger negative impact.         |

### Aerospace accuracy note

The guide intentionally describes the app as a preliminary analysis and visualization tool. Real lunar landing site certification would require orbital imagery, terrain-relative navigation data, lighting and shadow analysis over mission-specific time windows, plume-surface interaction assessment, communication link analysis, landing dispersion modeling, and formal verification.

# 5. LANDING ZONE SCORING LOGIC

The suitability score is normalized to a 0 to 100 scale. Higher scores indicate better combined performance across hazard clearance, slope, solar exposure, and communication coverage. The highest-scoring grid point becomes the recommended landing zone.

| Score Component     | Code Logic                                                      | Effect on LZ Selection                               |
|---------------------|-----------------------------------------------------------------|------------------------------------------------------|
| Hazard Score        | 1 minus weighted crater and boulder penalty.                    | Favors clear terrain away from obstacles.            |
| Slope Score         | 1 minus slope divided by 18 degrees, clamped to 0 to 1.         | Favors flatter touchdown zones.                      |
| Solar Score         | Elevation-normalized exposure proxy with a sinusoidal modifier. | Favors elevated areas with better modeled exposure.  |
| Communication Score | Relay-quality proxy with geometry and elevation terms.          | Favors terrain with better modeled relay visibility. |
| Aggregate Score     | Weighted sum divided by total active weights.                   | Balances mission priorities into one LZ score.       |

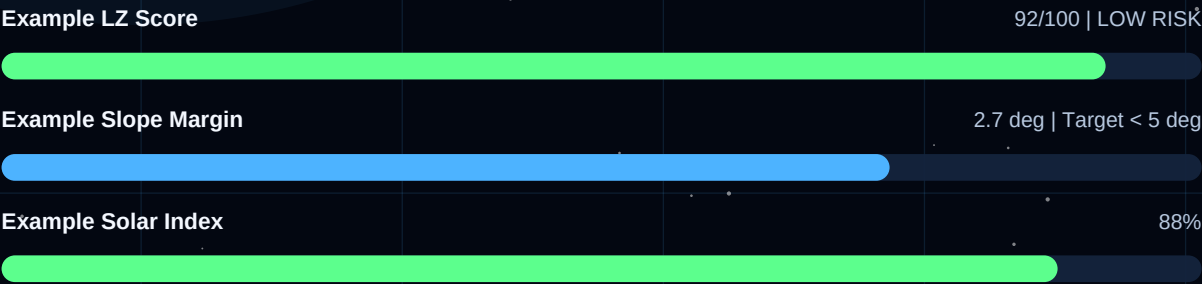
## Risk bands

| LZ Score  | Risk Band | Interpretation                                                                               |
|-----------|-----------|----------------------------------------------------------------------------------------------|
| 85 to 100 | LOW       | Strong candidate, assuming external verification confirms terrain conditions.                |
| 70 to 84  | MODERATE  | Usable candidate, but requires review of slope, hazards, lighting, and route constraints.    |
| Below 70  | HIGH      | Do not treat as a primary candidate without significant mitigation or alternate site review. |

# 6. DASHBOARD METRICS

The top metric row gives a fast mission-readiness snapshot. Review these values before interpreting detailed maps because a visually attractive site can still fail operationally due to slope, local obstacles, poor power geometry, or weak relay coverage.

| Metric             | Meaning                                                | Target / Interpretation                                          |
|--------------------|--------------------------------------------------------|------------------------------------------------------------------|
| Landing Zone Score | Aggregate suitability score from all scoring layers.   | Higher is better; risk band is shown beside the value.           |
| Slope at LZ        | Local slope at the recommended point.                  | The app labels target guidance as less than 5 degrees.           |
| Hazards Near LZ    | Count of craters and boulders near the candidate site. | Lower is better; zero or one is preferred for a clean candidate. |
| Solar Index        | Relative solar exposure proxy.                         | Higher supports power viability, especially for surface assets.  |
| Comm Index         | Relative communication coverage proxy.                 | Higher supports command, telemetry, and contingency operations.  |





# 7. 3D MAP AND ANALYSIS PANELS

The 3D terrain map is the main visual product. It colors the terrain by landing-zone score, overlays crater hazards, overlays boulder hazards, and marks the recommended LZ. The three secondary maps isolate slope, solar exposure, and communication coverage so users can diagnose why the site was selected.

| Panel                   | How to Use It                                                                                                 |
|-------------------------|---------------------------------------------------------------------------------------------------------------|
| 3D Terrain Map          | Inspect the recommended LZ in relation to terrain shape, crater fields, boulders, and broad elevation trends. |
| Slope Hazard Map        | Confirm that the LZ is not sitting on a sharp gradient or crater rim.                                         |
| Solar Exposure Analysis | Check whether the site sits in a favorable modeled exposure region.                                           |
| Communication Coverage  | Check whether the site has a favorable modeled relay geometry.                                                |
| Color Bars              | Use color scales for relative comparison, not absolute certification.                                         |

## Interpretation rule

Do not rely on the aggregate score alone. A high score should be supported by low slope, low local hazard count, and acceptable solar and communication values. If one layer is weak, adjust the mission weights and compare candidate behavior with the same simulation seed.

# 8. HABITAT PLANNING RECOMMENDATION

The habitat planning panel separates the touchdown point from the habitation area, power zone, and communication relay. This reflects a defensible surface-operations concept: the best touchdown point is not automatically the best place to establish crew or infrastructure assets.

| Element              | Placement Logic                                    | Rationale                                                                             |
|----------------------|----------------------------------------------------|---------------------------------------------------------------------------------------|
| Primary Landing Zone | Highest suitability point from the scoring engine. | Chosen for terrain safety and mission-weighted performance.                           |
| Habitat Zone         | Offset from the touchdown point.                   | Reduces exposure to descent plume effects, dust lofting, and landing dispersion risk. |
| Solar Power Zone     | Placed toward a modeled higher-exposure area.      | Supports surface power generation and energy resilience.                              |
| Comms Relay          | Placed to improve modeled communication geometry.  | Improves command, telemetry, and contingency operations.                              |

## Planning caution

The current offsets are simple geometric rules, not optimized site-layout solutions. In a mature mission design, these placements would be constrained by terrain traversability, line of sight, cable routing, dust contamination, emergency egress, thermal environment, and crew safety requirements.

# 9. ROVER PATH PLANNING

The rover planning module generates a closed survey route around the selected landing zone. It uses the selected survey radius, waypoint count, and rover speed to estimate route length and survey duration.

| Input                 | Function                                   | Best Use                                                                                 |
|-----------------------|--------------------------------------------|------------------------------------------------------------------------------------------|
| Rover Survey Speed    | Sets traversal speed for time calculation. | Use conservative values for rough terrain demonstrations.                                |
| Survey Radius         | Controls the area inspected around the LZ. | Increase for habitat preparation, reduce for quick site survey.                          |
| Survey Waypoints      | Controls route complexity.                 | More waypoints create a denser inspection pattern.                                       |
| Survey Path Length    | Total route distance.                      | Use as a planning proxy for rover energy and timeline.                                   |
| Estimated Survey Time | Path distance divided by rover speed.      | Use for comparison only; real traverse time requires terrain and operations constraints. |

## Route interpretation

The path is a planning abstraction. It should be treated as an initial survey pattern for scouting hazards, validating terrain assumptions, and selecting final habitat-support locations.

# 10. MISSION EXPORT WORKFLOW

The export section packages the mission state into JSON and CSV. JSON preserves nested information such as recommended LZ coordinates and recommendation text. CSV is better for spreadsheet review, quick comparisons, or traceability matrices.

| Export              | Contains                                                                                                                               | Use Case                                                                   |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Mission Report JSON | Mission ID, target body, profile, timestamp, LZ coordinates, scores, hazard counts, survey distance, survey time, and recommendations. | Best for technical archiving or downstream app integration.                |
| Mission Summary CSV | One-row mission summary with scalar values.                                                                                            | Best for class reports, spreadsheet comparison, and simple dashboard logs. |

## Recommended file discipline

- Keep the mission ID in the exported file name for traceability.
- Record the simulation seed whenever a screenshot or score is used in a report.
- Do not compare two scenarios unless the terrain seed and primary model settings are controlled.
- Use JSON when recommendations and nested mission objects must be preserved.
- Use CSV when building a trade-study table across many runs.

# 11. ENGINEERING LIMITS AND BEST PRACTICES

This app is a strong professional prototype for mission visualization, autonomy workflow demonstration, and systems engineering communication. It should not be represented as a validated flight, landing, or site-certification tool.

| Area                | Limit                                                              | Best Practice                                                               |
|---------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Terrain Fidelity    | The map is procedurally generated.                                 | Label results as simulated unless tied to real terrain data.                |
| Solar Model         | Solar exposure is a relative proxy.                                | Use actual illumination and shadow modeling for mission planning.           |
| Communication Model | Relay coverage is a simplified proxy.                              | Use link-budget and line-of-sight analysis for real missions.               |
| Hazard Detection    | Craters and boulders are generated objects, not sensor detections. | Describe the module as hazard-modeling or detection-inspired analysis.      |
| Landing Safety      | No descent dynamics or dispersion ellipse is modeled.              | Add landing ellipse, TRN uncertainty, and plume effects in future versions. |
| Rover Planning      | Path ignores traction, slopes, obstacles, and energy.              | Add terrain-cost routing for a more realistic rover module.                 |

## Suggested next upgrades

- Add real digital elevation model import for lunar sites.
- Add landing ellipse visualization and keep-out zones.
- Add a terrain-cost path planner for rover traverse safety.
- Add illumination windows based on mission date and local topography.
- Add a requirements traceability page linking each output to mission needs.

## Final positioning statement

Position the app as a polished autonomy and systems engineering demonstrator for landing-zone assessment, not as a certified lunar landing system. That framing is accurate, defensible, and professional.